ARSLAN ARSHAD

Senior AI & Backend Engineer | Python, FastAPI, Django | LangGraph, RAG & LLM Systems

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Fully Remote | US Eastern & EU Central | Immediate Availability

Professional Summary

Senior AI and backend engineer with 6+ years building production-grade Python systems with FastAPI and Django, specializing in LLM applications, RAG pipelines, and agentic multi-agent orchestration with LangGraph and LangChain. Leads system design and mentors engineers on scalable infrastructure. Proven track record: collapsed 4-hour research workflows to 10 minutes; delivered an estimated 35–40% improvement in API performance; scaled multi-tenant SaaS platforms handling complex LLM workloads reliably and at scale.

Technical Skills

Machine Learning & LLM Systems: LangGraph, LangChain, RAG architecture, multi-agent orchestration, agentic workflows, semantic search, vector databases (Pinecone, FAISS), LLM orchestration, prompt engineering, LLaMA, Gemini, OpenAI APIs
Backend & Systems: Python, FastAPI, Django, Django REST Framework, REST APIs, GraphQL, WebSockets, async processing (Celery/Redis), clean architecture, microservices
Data & Performance: PostgreSQL, MySQL, MongoDB, Redis, SQLite, query optimization, indexing strategy, connection pooling, caching patterns
Infrastructure & Security: AWS (EC2, RDS, S3), Docker, CI/CD, production monitoring, JWT, OAuth 2.0, RBAC, tenant isolation
Integrations: Stripe (payments), Plaid (banking), Twilio (communications)
Frontend (Secondary): ReactJS, JavaScript, HTML5, CSS3

Professional Experience

Senior Software Engineer, AI & Backend Architecture Dec 2024 – Present
Kinetic Intelligence *Remote (Dubai, UAE)*

- Lead architecture and end-to-end delivery across a distributed backend of **4 microservices** (Django/DRF and FastAPI) plus the **AI Coach service**, defining technical direction and cross-service design decisions for an AI-powered athlete development platform.
- Architected **multi-agent conversational AI services** for coaching chat, scouting reports, and box-score analysis, each with session-based context and follow-up chat continuity.
- Built an automated **daily reports pipeline** pairing an MTR scoring engine (Move/Train/Recover) with LLM-generated recommendations, run by **Celery/Redis** with an hourly scheduler and historical backfill jobs.
- Designed **vision-LLM document intelligence workflows** to extract and classify scouting/box-score data and auto-verify tax-exemption certificates and purchase orders, eliminating manual review.
- Own the **Users and Notifications services** (Django/DRF) — JWT multi-profile auth, Stripe/Firebase/Twilio integrations, and an FCM/APNS device pipeline coordinated across services via async Celery task fan-out on login, logout, and profile switch.
- Delivered **40% reduction in API response times** by architecting Redis as a dual-purpose cache and Celery broker, migrating the **training service** off APScheduler to Celery+Redis+Beat, and resolving a production DB connection-pool exhaustion via split API/worker/beat Kubernetes deployments.
- Establish engineering standards through systematic PR reviews, clean-architecture practices, and reusable pattern standardization; lead system design reviews and sprint planning in a distributed, async-first environment.

Full-Stack Software Engineer, AI & Data Systems Jan 2023 – Dec 2024
Inovaqo *Lahore, Pakistan*

- Engineered production multi-agent AI research system using LangGraph with autonomous planning, retrieval, tool orchestration, and structured reporting. Reduced manual market research workflows from 4+ hours to under 10 minutes. Deployed across 8+ enterprise clients.
- Built end-to-end RAG pipeline using LangChain and Pinecone for semantic search across large-scale document corpora. Improved search relevance and reduced query latency from 3.2s to 800ms through semantic chunking and intelligent context ranking.
- Improved API performance by an estimated 35% across high-traffic endpoints through systematic profiling, selective indexing, Celery/Redis async processing, and ORM optimization eliminating N+1 query patterns, supporting higher concurrent usage without additional infrastructure investment.
- Implemented scalable multi-tenant SaaS backend with database-level tenant isolation and tenant-scoped RBAC. Integrated Stripe, Plaid, and Twilio across 12+ payment and verification flows, with JWT/OAuth2 authentication.

Software Engineer Nov 2019 – Dec 2022
CodingCops *Lahore, Pakistan*

- Built and shipped 8+ full-stack web and mobile applications using Django REST Framework and ReactJS, owning complete feature lifecycle from requirements through production deployment across e-commerce, SaaS, and marketplace domains.
- Designed normalized PostgreSQL and MySQL schemas optimized for high-traffic patterns using query analysis, indexing strategies, and intelligent caching. Established reusable REST API contracts and React component libraries adopted across concurrent client projects.
- Conducted peer code reviews, contributed to sprint planning and architecture decisions, and established engineering best practices around testing, documentation, and code quality, adopted across the growing team.

Projects

AI-Powered Market Research Platform *FastAPI | React | LangGraph | Pinecone | Groq | LLaMA*

- **Problem:** Manual market research analysis takes 4+ hours per request, blocking enterprise clients from scaling operations.
- **Architecture:** Multi-agent LLM orchestration using LangGraph (planner, researcher, generator agents). RAG infrastructure with Pinecone, semantic chunking, and LLM-based evaluation loops. Confidence scoring with automatic query rewriting reduces hallucinations.
- **Result:** Automated workflow completes in 10 minutes, deployed across 8+ enterprise clients.

YouTube Talker – AI Video Content Analyst *Flask | LangChain | Gemini | FAISS | YouTube API*

- **Problem:** Content creators spend 45+ minutes manually reviewing long-form videos (1–6+ hours) to extract insights.
- **Architecture:** End-to-end GenAI pipeline extracting transcripts via YouTube API, generating embeddings with LangChain, indexing with FAISS. Context-aware answers using Gemini grounded in retrieved chunks with source attribution.
- **Result:** Content analysis time reduced from 45 minutes to 30 seconds per video.

InsightRAG – Self-Correcting Document Q&A Agent *FastAPI | LangGraph | LangChain | FAISS | Groq*

- **Problem:** Document Q&A systems hallucinate and lack transparent error handling—problematic for compliance-critical workflows.
- **Architecture:** Self-grading RAG agent using LangGraph: rewrites ambiguous queries, retrieves documents, grades context relevance, and runs automatic refinement loops. Multi-signal confidence scoring combines vector similarity, lexical overlap, and LLM-based assessment.
- **Result:** Uncertain cases route to human review with fallback mechanisms, reducing false positives on sensitive data.

Education

Bachelor of Science in Software Engineering
Punjab University College of Information and Technology (PUCIT) *Lahore, Pakistan*